

Answers to 2020 Final Exam (Optional Questions)

13. $\frac{1}{2}(\ln 2)^2$

14. $f(x) = \frac{\frac{10}{11}e^x}{1 - \frac{10}{11}e^x}, \quad f(x) = \frac{-\frac{10}{11}e^x}{1 - \frac{10}{11}e^x}, \quad f(x) = \frac{\frac{10}{9}e^x}{\frac{10}{9}e^x - 1}, \quad f(x) = \frac{-\frac{10}{9}e^x}{\frac{10}{9}e^x - 1}$

15. $y(t) = C_1 e^{\frac{1}{t}} + C_2 t e^{\frac{1}{t}}$

16. $N - 1$ ($N - 1$ fold) and $2N - 1$